

ARAR, Saudi Arabia

WMO: 403570

Lat: 30.907N

Lon: 41.138E

Elev: 553

StdP: 94.86

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.2	-8.9	1.9	8.5	-6.8	2.3	12.7	12.2	11.6	10.8	12.4	2.7	280	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.8	45.2	23.5	44.0	22.9	42.9	22.3	25.0	42.3	24.1	41.5	23.3	40.3	3.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.8	30.5	18.9	14.7	30.5	17.9	13.8	30.7	79.4	42.3	75.8	41.7	72.6	40.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.4	8.4	-1.9	47.0	2.0	1.2	-3.4	47.9	-4.6	48.6	-5.7	49.2	-7.2	50.1		
			-3.4	24.8	2.2	2.3	-4.9	26.5	-6.2	27.8	-7.4	29.1	-9.0	30.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	10.4	12.8	17.3	22.7	28.2	32.4	34.7	34.9	31.3	25.6	16.7	11.6
	DBStd	9.33	2.72	3.37	3.68	3.63	3.22	2.14	2.11	2.06	2.55	3.40	3.77	3.18
	HDD10.0	58	26	11	1	0	0	0	0	0	0	0	2	19
	HDD18.3	758	245	158	64	6	0	0	0	0	0	1	74	210
	CDD10.0	4905	40	89	227	382	566	671	765	771	640	484	203	68
	CDD18.3	2561	0	2	32	137	307	421	507	512	390	227	25	1
	CDH23.3	39502	4	45	414	1657	4344	6812	8702	8753	5881	2633	242	14
	CDH26.7	26127	0	7	138	775	2621	4628	6277	6328	3870	1415	66	1
(14) Wind	WSAvg	4.0	3.7	4.1	4.5	4.5	4.4	4.5	4.5	3.9	3.5	3.5	3.3	3.3
(15) Precipitation	PrecAvg	49	9	7	5	7	2	0	0	0	0	4	7	8
	PrecMax	144	22	50	34	59	17	0	1	0	0	26	43	55
	PrecMin	9	1	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	28	5	10	8	13	5	0	0	0	0	6	9	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	27.2	34.2	37.2	41.3	45.0	46.8	46.9	44.2	39.2	31.2	25.1
		MCWB	14.1	15.0	18.0	18.8	21.1	22.9	24.5	24.6	23.1	21.4	18.2	14.1
	2%	DB	20.7	24.2	30.0	35.1	39.6	43.0	45.2	45.2	42.8	37.2	28.2	22.1
		MCWB	12.9	14.5	16.6	18.1	20.3	22.2	23.5	23.7	22.8	19.6	16.9	13.5
	5%	DB	18.2	22.1	27.8	33.1	38.1	41.8	44.0	44.1	41.1	35.9	26.0	19.9
		MCWB	12.0	13.4	15.7	17.8	19.7	21.3	22.7	23.1	21.7	19.0	16.3	13.5
	10%	DB	16.8	19.9	25.2	31.1	36.7	40.2	42.8	42.9	39.2	33.9	23.9	18.0
		MCWB	11.3	12.7	14.5	17.1	19.3	20.4	21.9	22.5	20.6	18.9	15.7	12.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	18.3	20.3	22.2	23.7	24.6	25.7	26.2	25.2	23.2	20.8	18.1
		MCDB	19.3	23.1	27.0	31.7	36.2	42.2	43.4	43.4	41.3	32.5	24.7	21.3
	2%	WB	14.2	16.1	18.3	20.5	22.2	23.4	24.6	25.2	23.8	21.9	19.2	15.8
		MCDB	18.2	21.6	27.2	29.9	35.9	40.6	42.3	43.1	39.7	32.9	24.9	18.8
	5%	WB	13.0	14.5	16.9	19.3	21.1	22.4	23.8	24.3	22.8	20.6	17.8	14.4
		MCDB	16.9	19.8	25.2	28.5	34.7	39.4	42.0	42.2	38.9	31.3	23.3	18.0
	10%	WB	11.9	13.2	15.5	18.1	20.2	21.6	23.0	23.3	21.9	19.7	16.7	13.2
		MCDB	16.1	18.8	23.6	28.7	33.5	38.4	41.0	41.0	37.6	30.6	21.7	17.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.8	13.8	14.4	14.6	15.3	15.4	15.8	15.6	14.4	12.6	12.3
		MCDBR	14.3	15.4	16.2	16.2	15.8	16.6	16.4	16.6	16.9	16.3	14.7	14.0
		MCWBR	8.4	8.4	8.3	7.4	6.8	7.2	7.3	7.4	7.6	7.2	7.6	8.6
	5% WB	MCDBR	12.4	13.2	14.2	14.2	15.0	15.7	15.8	16.4	15.7	14.5	12.0	11.8
		MCWBR	8.1	8.0	8.0	7.2	6.9	7.2	7.4	7.6	7.6	6.9	6.7	8.1
(40) Clear-Sky Solar Irradiance	taub		0.369	0.401	0.457	0.544	0.559	0.489	0.467	0.484	0.464	0.495	0.425	0.383
	taud		2.092	1.993	1.840	1.671	1.657	1.811	1.887	1.848	1.876	1.812	1.953	2.051
	Ebn at Noon		834	843	821	766	757	809	825	805	804	735	760	791
	Edh at Noon		132	160	201	247	252	215	199	204	191	190	151	131
(44) All-Sky Solar Radiation	RadAvg		3.61	4.60	5.79	6.64	7.40	8.21	7.89	7.26	6.27	4.90	3.81	3.33
	RadStd		0.16	0.25	0.18	0.23	0.29	0.17	0.24	0.17	0.21	0.22	0.30	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+1.39	N/A	N/A	+1.10	N/A	N/A	-32	-143	+462	+306

Nomenclature: See separate page